J. Fluid Mech. (2026), *vol.* 1032, R2, doi:10.1017/jfm.2026.11411



Routes towards homogeneity in stratified turbulent flows

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(Received 26 January 2026; revised 5 March 2026; accepted 5 March 2026)

Quantifying the rate at which a stratified turbulent flow mixes a density field is of crucial importance for many environmental and industrial applications. In the absence of molecular diffusion κ (i.e. in the absence of irreversible mixing), a stratified turbulent flow forced so as to have a constant kinetic energy will converge towards a statistical steady state whose density field geometric properties depend on the Richardson number Ri (defined as the ratio of the kinetic energy in the flow to the amount of energy required to overturn the full water column). This statistical steady state is reached after vertically disturbed fluid parcels have explored the depth that is accessible energetically and have returned to their neutrally buoyant position, i.e. after a 'resetting time' t_R. The magnitude of t_R is controlled by stratification strength N and the buoyancy Reynolds number Re_b, quantifying the ratio between the Kolmogorov and Ozmidov scales, and hence the range of scales effectively unaffected by stratification. When $\kappa \neq 0$, a second time scale needs to be considered: the mixing time scale t_M. Within a mixing time, diffusion smooths the density field. We show that the ratio of the mixing and resetting times t_M/t_R, as well as Ri, control how fast stratified turbulent flows mix a density field into a fully homogeneous state and, hence, the history of mixing in such flows. In particular, we identify three regions in the $(Ri, t_M/t_R)$ parameter space for which the time evolution of measures of mixing is controlled by different algebraic combinations of t_M/t_R and Ri. These scaling laws are compared with idealised direct numerical simulations. Using these findings, we propose a simple model for the time evolution of the density histogram in stratified turbulent flows.

Key words: turbulent mixing, stratified flows

1. Introduction

Mixing, i.e. the way in which a scalar field irreversibly evolves from a segregated state towards homogeneity, is a key process in a large variety of settings with length and time

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scales spanning orders of magnitude. Of particular significance is mixing of density in oceanic flows, where the rate at which turbulence mixes the water column is known to have a leading order impact on the large-scale global ocean circulation (Wunsch & Ferrari 2004), biological productivity (Marra, Bidigare & Dickey 1990) and tracer pathways (Cimoli *et al.* 2023). It is currently impossible to resolve mixing directly in observational measurements and large-scale models, and approximate mixing proxies and parametrisations are hence required.

Mixing is inherently related to the time evolution of the probability density function (or histogram) of the scalar being mixed (Villermaux 2019). The scalar at hand in this work is the density; we will denote $p(\rho)$ the density histogram, with ρ the observed density levels. Since stirring reorganises the scalar field without modifying its content, the histogram p can only vary due to molecular mixing. In particular, in the absence of scalar sources, mixing will bring the scalar field towards a fully homogeneous state characterised by $p(\rho) = \delta(\rho - \langle\rho\rangle)$ (with δ the Dirac measure and $\langle\rho\rangle$ the average of the initially present density levels). Therefore, quantifying mixing requires understanding the time-evolution of the scalar histogram and the parameters that control it. The density histogram can be used to infer mixing proxies, such as the background potential energy $\mathcal{E}_b$ (i.e the minimum potential energy of the system which can be accessed through adiabatic rearrangement of fluid elements; Tseng & Ferziger (2001)) and hence the cumulative mixing efficiency (Caulfield 2021)

$$\eta := \frac{\mathcal{E}_b - \mathcal{E}_b(t=0)}{\mathcal{E}_b - \mathcal{E}_b(t=0) + \mathcal{E}_k}, \tag{1.1}$$

with $\mathcal{E}_k$ the turbulent kinetic energy.

In the case of passive scalar mixing (i.e. for which the scalar field does not affect the velocity field stirring it, such as mixing dye in a turbulent flow), models have been suggested for the time evolution of the scalar histogram (Villermaux 2019). In particular, it has been shown that mixing acts on the mixing time scale $t_M \sim \gamma^{-1}\mathcal{F}(Pr)$ with γ a measure of the stretching rates deforming the small-scale structures in the scalar field into thin 'lamellae' (i.e. the stirring) over which molecular diffusion can act efficiently. The function $\mathcal{F}(Pr)$ is a weak correction (typically logarithmic) in the Prandtl number $Pr := \nu/\kappa$ (with κ and ν the molecular diffusivity and viscosity, respectively) (Duplat, Innocenti & Villermaux 2010). It is usual to estimate the (average) stretching rate γ as $\sqrt{\epsilon/\nu}$, with ϵ the dissipation rate of turbulent kinetic energy (Batchelor 1959). For the dynamically active case of mixing density (which is coupled to the velocity field stirring it), such a model is (to our knowledge) as yet developed. However, experimental studies have shown that mixing time scales can still be used to infer the efficiency of mixing in simple (laminar) flow geometries (Petropoulos *et al.* 2023). In this work, we aim to extend this framework to stratified turbulent flows and ask the following questions. In the absence of density sources, what flow parameters determine the history of mixing in stratified turbulent flow? How fast does a density field stirred by a turbulent flow evolve from an initially segregated configuration to a fully homogeneous state? In particular, we aim to describe the different time scales that govern the time evolution of $p(\rho)$ and hence η. Our findings are used to develop a simple model for $p(\rho)$ and are compared with idealised direct numerical simulations.

2. Phenomenology

To gain insight into the problem, it is relevant to consider the prototypical case where the stirring velocity field has a constant kinetic energy $\mathcal{E}_k$ (per unit mass). In that case,

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the long-time limit of the cumulative mixing efficiency η only depends on $\mathcal{E}_k$ and the initial density field configuration. For instance, the background potential energy (per unit volume, scaled by a reference density ρ_0) for an initially linearly stratified density field $\rho(z, t=0) = -N^2\rho_0 z/g$ (with z the vertical coordinate, N the buoyancy frequency and g the acceleration of gravity) and a fully mixed final state (characterised by a single density level $\langle\rho\rangle := \int_0^H \rho(z, t=0)\mathrm{d}z/H = -N^2\rho_0 H/(2g)$ with H the depth of the water column) is

$$\mathcal{E}_b(t=0) = \frac{g}{\rho_0 H}\int_0^H \rho(z, t=0) z \,\mathrm{d}z = -\frac{1}{3}N^2H^2, \tag{2.1}$$

and

$$\mathcal{E}_b(t\to+\infty) = \frac{g}{\rho_0 H}\int_0^H \langle\rho\rangle z \,\mathrm{d}z = -\frac{1}{4}N^2H^2, \tag{2.2}$$

respectively. Hence,

$$\eta(t\to+\infty) = \frac{1/12}{1/12 + 1/Ri} \quad \text{with} \quad Ri = \frac{N^2H^2}{\mathcal{E}_k}. \tag{2.3}$$

In the two-layer configuration depicted in figure 1, a similar computation gives

$$\eta(t\to+\infty) = \frac{1/8}{1/8 + 1/Ri} \quad \text{with} \quad Ri = \frac{gH}{\mathcal{E}_k}\times\frac{\rho_0-\rho_1}{\rho_0}. \tag{2.4}$$

The above-mentioned long-time limits of the mixing efficiency η are decreasing functions of the $\mathcal{E}_k$: the more energy needing to be input into the system to fully homogeneise it, the lower the mixing efficiency. Note that for $Ri < 1$, the kinetic energy in the flow is sufficient to overturn the full water column; for $Ri > 1$, it is not. Equation (2.4) justified the constant kinetic energy scenario; the long-time limit of the mixing efficiency is fixed and we here focus on its time evolution towards that limit.

Two processes control the time evolution of η towards its long-time limit: stirring that reorganises the density field without changing η, and mixing that smooths the density field, increases the background potential energy $\mathcal{E}_b$ and hence η. We study these two processes in isolation and then in combination.

In the absence of mixing ($\kappa = 0$), turbulent fluctuations (stirring) and buoyancy forces (restratification) compete so that the vertical displacement of fluid parcels with a given density content (that is constant since $\kappa = 0$ for now) is constrained (Kimura & Herring 1996). In the initially linearly stratified configuration, the furthest (dimensionless) vertical distance such fluid parcels can travel away from their initial position is $L_S/H \sim \sqrt{\mathcal{E}_k/(N^2H^2)} = \sqrt{1/Ri}$ (a result that arises from a force balance between inertia and buoyancy on fluid parcels; see Petropoulos, de Bruyn Kops & Caulfield (2025)). In the two-layer configuration of figure 1, we have $L_S/H \sim 1/Ri$ (Venaille *et al.* 2017). The weaker the stratification, or the more energy there is in the system, the further away fluid parcels can vertically travel. This has important implications for mixing, as the 'screening length' L_S inevitably constrains the density levels that fluid parcels encounter along their stochastic path and hence mix into; for $L_S/H \ll 1$ (i.e. $Ri \gg 1$), the density levels seen by fluid parcels are close to the density they carry and we expect the route to homogeniety to be slow. By stirring the density field with constant kinetic energy, we fix this parameter.

As a consequence of the competition between stirring and restratification, in the diffusiveless limit $\kappa = 0$, the probability $p(\rho, z)$ of finding a density level ρ at depth z reaches (maximum disorder) a steady-state limit (Venaille *et al.* 2017) characterised by

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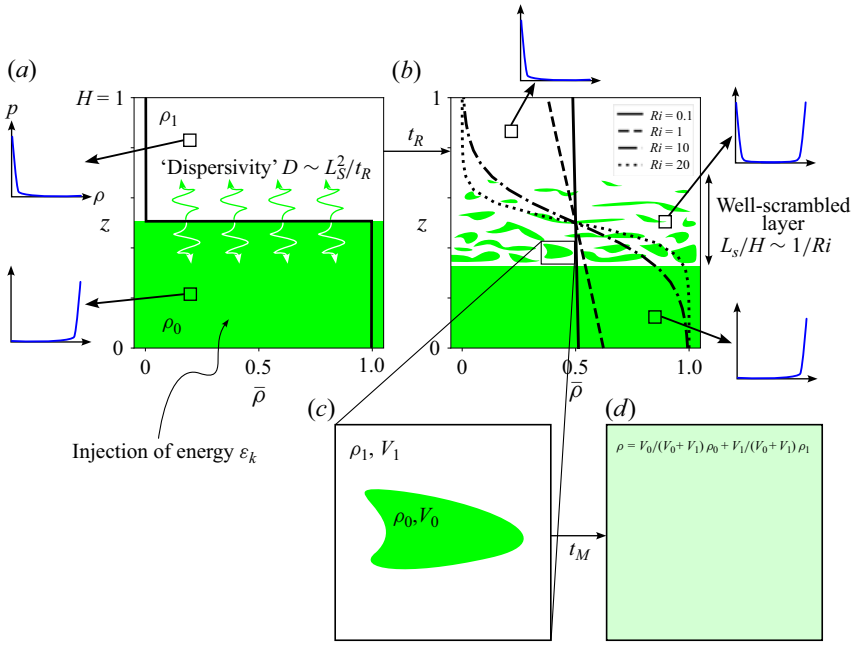


Figure 1. Evolution of a two-layer density field in the absence (top) or presence (bottom) of mixing. The density level ρ_0 and $\rho_1 < \rho_0$ are represented in green and white, respectively; shades of green correspond to intermediate density levels. As turbulent fluctuations act against stratification to stir the box and vertically disperse fluid parcels with different densities, they create a steady-state density field characterised by a well-scrambled region where dense and light fluid coexist ($(a) \rightarrow (b)$). This steady state is reached within a time t_R. Thick black lines in panel (b) correspond to the steady-state horizontally averaged density field derived by Venaille, Gostiaux & Sommeria (2017) ($\overline{\rho} \propto \tanh(Riz/H)$) for various values of the Richardson number Ri. Note that in the two-layer configuration, the well-scrambled region has a thickness $L_S/H \sim 1/Ri$ (rather than $L_S/H \sim 1/\sqrt{Ri}$ as in the linearly stratified case). The blue curves correspond to the probability of finding a given density level at a given point in space $p(\rho, z)$. The effect of mixing is to smooth density differences between neighbouring fluid parcels ($(c) \rightarrow (d)$) within a time t_M. Here, we model the effect of mixing as an averaging process in density space, i.e. a (weighted) convolution in probability space. In this work, we will assume that the volumes of the fluid parcels mixing into each other are equal ($V_0 = V_1$) so that $\alpha := V_0/(V_0 + V_1) = 1/2$.

'well-scrambled' (or 'overturned') regions of dimensionless depth L_S/H (figure 1). We hypothesise here that the characteristic time scale to reach this equilibrium state (i.e. to overturn layers of depth L_S) is the time for a fluid parcel to come back to its equilibrium position due to restoring buoyancy forces after being vertically disturbed. Because velocity gradients typically stretch fluid parcels into thin and elongated structures, or lamellae, with characteristic thickness below the scale at which viscosity starts to matter (Duplat *et al.* 2010; Villermaux 2019), this time is not the inverse of the buoyancy frequency, but rather the resetting time $t_R \sim Re_b/\gamma \sim \sqrt{Re_b}/N$ with γ the average stretching rate, and $Re_b := \epsilon/(\nu N^2)$ the buoyancy Reynolds number and ϵ the dissipation rate of turbulent kinetic energy (Petropoulos *et al.* 2025). Here, we used the estimate $\gamma \sim \sqrt{\epsilon/\nu}$ for the (average) stretching rate (Batchelor 1959). The viscous correction is crucial. It takes into account that the settling dynamics of the lamellae towards their neutrally buoyant positions is dictated by a balance between viscous drag and buoyancy forces due to the small characteristic scales involved (Petropoulos *et al.* 2023, 2025). Note that in the two-layer configuration of figure 1, this balance gives $t_R \sim \gamma \mathcal{E}_k \rho_0^2/[g(\rho_0 - \rho_1)]^2 \sim \sqrt{\epsilon/\nu} \times \mathcal{E}_k \rho_0^2/[g(\rho_0 - \rho_1)]^2$.

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The screening length scale L_S and resetting time t_R can be used to define a 'dispersivity' $D := L_S^2/t_R$ characterising the dispersive effect of turbulent fluctuations in the presence of stratification (in the diffusiveless limit $\kappa = 0$).

We now consider the effect of mixing ($\kappa \neq 0$). As turbulent fluctuations disorder the density field, small-scale density structures (the lamellae) with different density content will encounter each other and mix; this happens on the mixing time scale $t_M \sim \gamma^{-1}\mathcal{F}(Pr) \sim \sqrt{\nu/\epsilon}\mathcal{F}(Pr)$ (Petropoulos *et al.* 2023, 2025). Then, if $t_M < t_R$, we expect η to reach its long-time limit relatively fast; as soon as parcels are vertically disturbed, they mix with their surroundings. Conversely, if $t_M > t_R$, turbulence first stirs the density field towards its maximum disorder equilibrium state before mixing and the time evolution of η is slower; in particular, mixing is not efficient in the sense that vertically disturbed fluid parcels do not have time to mix before returning to their neutrally buoyant position. More precisely, we hypothesise the existence of the following regimes in the $(Ri, t_M/t_R)$ parameter space (summarised in figure 2).

(i) Full overturn of the water column, i.e. $Ri < 1$. In that case, there is enough energy to overturn the full water column ($L_S > H$) and, hence, for fluid parcels to potentially encounter all the density levels present in the column along their stochastic paths. If the mixing time is larger than the time for turbulent fluctuations to overturn the entire water column (i.e. $t_M > H^2/D$ or, equivalently, $Ri < t_M/t_R$), mixing is mixing-limited. We expect the water column to be fully mixed after a time of order $t_R + t_M$, i.e. when scaled by t_R, of order t_M/t_R in the limit $t_M \gg t_R$. The turbulent fluctuations first overturn the water column, then mixing happens at once. In contrast, if $t_M < H^2/D$, mixing of the water column is limited by dispersion. Mixing happens before turbulent fluctuations have time to overturn the water column. We expect that, after a mixing time t_M, well-mixed layers of depth $\sqrt{Dt_M}$ are formed (this corresponds to the vertical excursion of fluid parcels due to the dispersivity D within a mixing time). There are $H/\sqrt{Dt_M}$ such layers to mix with each other. Hence, the time to fully mix the water column is expected to be of order $t_M \times H/\sqrt{Dt_M}$, i.e. $\sqrt{Ri \times t_M/t_R}$ when scaled by t_R.

(ii) Partial overturn of the water column, i.e. $Ri > 1$. In that case, the energy in the system can only overturn layers of depth $L_S < H$. In the case where mixing happens before turbulent fluctuations can overturn the depth L_S of the water column that is accessible energetically (i.e. $t_M < L_S^2/D$ or, equivalently, $t_M/t_R < 1$), mixing is dispersion limited. The above-mentioned argument holds and we expect the water column to be fully mixed after a (dimensionless) time of order $\sqrt{Ri \times t_M/t_R}$. In contrast, if $t_M/t_R > 1$, we expect that after t_R, there are H/L_S 'well-scrambled' layers (in that regime, mixing has no time to act before fluctuations overturn layers of depth L_S) that need to be mixed internally and with each other. Hence, the time to mix the entire water column is expected to be of order $t_M \times H/L_S$, or when scaled by t_R, $\sqrt{Ri} \times t_M/t_R$.

The different elements described previously can be summarised into the following equation for the evolution of the probability $p(\rho, z)$ of finding a density level ρ at depth z:

$$\partial_t p = \underbrace{D\partial_z^2 p}_{\text{dispersion}} + \underbrace{\frac{Dg}{\beta\rho_0}\partial_z[(\rho - \overline{\rho})p]}_{\text{restratification}} + \frac{1}{t_M}\underbrace{(-p + p_\alpha * p_{1-\alpha})}_{\text{mixing}}. \tag{2.5}$$

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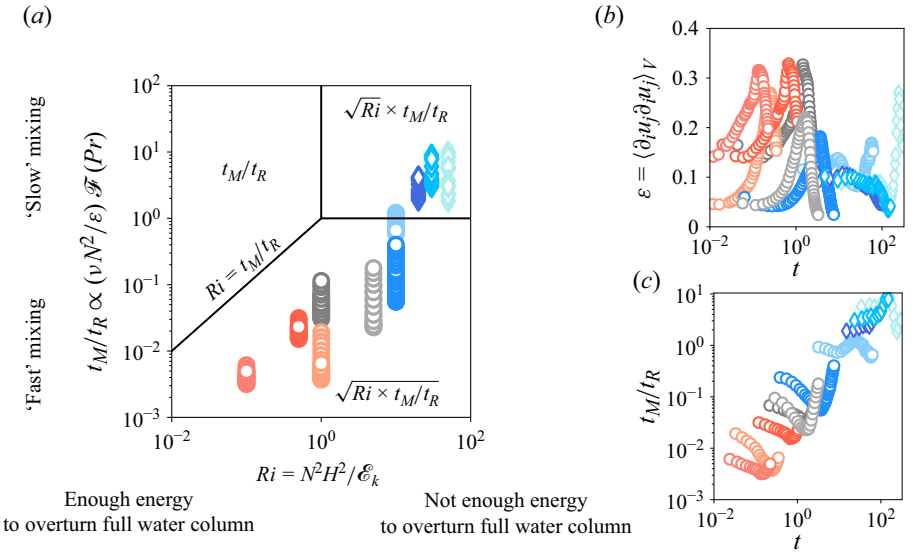


Figure 2. (*a*) Regime diagram in $(Ri, t_M/t_R)$ space. (*b*) Time evolution of the dissipation rate of turbulent kinetic energy ϵ. (*c*) Time evolution of the ratio of time scales t_M/t_R. The (dimensionless) time scale governing how fast the water column is fully mixed is presented as text insets. Symbols correspond to simulation data: $Ri_L = 0.1,\ H = 1$; $Ri_L = 0.5,\ H = 1$; $Ri_L = 0.1,\ H = \sqrt{10}$; $Ri_L = 1,\ H = 1$; $Ri_L = 0.5,\ H = \sqrt{10}$; $Ri_L = 1,\ H = \sqrt{10}$; $Ri_L = 10,\ H = 10$; $Ri_L = 20,\ H = 1$; $Ri_L = 30,\ H = 1$ and $Ri_L = 50,\ H = 1$.

Here, $\overline{\rho}(z) := \int \rho p(\rho, z)\,\mathrm{d}\rho$ is the instantaneous mean density profile and $p_\alpha(\rho, z) = p(\rho/\alpha, z)/\alpha$. The convolution operation $*$ is in density space.

The dispersion and restratification parts of (2.5) follow from work by Venaille *et al.* (2017). The constant β is chosen so that, in the diffusiveless limit $t_M/t_R \to +\infty$, (2.6) has a steady-state solution that exhibits a characteristic length scale L_S, i.e. $D/L_S^2 \sim Dg/(\beta L_S)$. The dispersion operation models turbulent fluctuations that broaden the distribution of density levels found at a given height z. Restratification towards the instantaneous mean density profile $\overline{\rho}$ models the effect of buoyancy that increases the probability of finding a fluid parcel close to its neutrally buoyant (with respect to $\overline{\rho}$) position. Mixing is modelled as a convolution operation, translating the fact that mixing averages the density of neighbouring fluid parcels within a mixing time (figure 1). Importantly, by construction, this operation preserves the average density and its stationary solution is the Dirac measure $\delta(\rho - \langle\rho\rangle)$ (see Appendix A). Here, we assume that the weight $\alpha = 1/2$, i.e. that the different fluid structures mixing into each other have approximately the same volume.

When scaling time t by the resetting time t_R, height z by the depth of the water column H and the density ρ by the reference density ρ_0, (2.5) reads

$$\partial_t p = Ri^{-s}\partial_z^2 p + Ri^{-s/2}\partial_z[(\rho - \overline{\rho})p] + \frac{t_R}{t_M}(-p + p_\alpha * p_{1-\alpha}). \tag{2.6}$$

In the linearly stratified configuration, $s = 1$ and $t_M/t_R \sim \nu N^2 \epsilon^{-1}\mathcal{F}(Pr)$. In the two-layer system depicted in figure 1, $s = 2$ and $t_M/t_R \sim \nu g^2(\rho_0 - \rho_1)^2/(\epsilon \mathcal{E}_k \rho_0^2)\mathcal{F}(Pr)$. Note that mass conservation is ensured by the no-flux boundary conditions $Ri^{-s}\partial_z p + Ri^{-s/2}(\rho - \overline{\rho})p = 0$ for all ρ at $z = 0, 1$ (see Appendix A). This boundary condition and (2.6) model the time evolution of p through stirring and mixing. The background potential

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energy and, hence, the mixing efficiency η can then be estimated from p using the method of Tseng & Ferziger (2001).

3. Comparison with direct numerical simulations

We now compare the theoretical scalings described previously with three-dimensional direct numerical simulations. In this section, dimensional quantities are denoted by a hat $\hat{\cdot}$. Dimensionless quantities are left unchanged. We use the spectral element solver Dedalus (Burns *et al.* 2020) to solve the (dimensionless) incompressible Navier–Stokes equation under the Boussinesq approximation

$$\begin{cases} \partial_t \boldsymbol{u} + \boldsymbol{u}\cdot\boldsymbol{\nabla}\boldsymbol{u} = -\boldsymbol{\nabla} p + Re_L^{-1}\nabla^2\boldsymbol{u} - Ri_L\rho\boldsymbol{e}_z + A\boldsymbol{u}, \quad \boldsymbol{\nabla}\cdot\boldsymbol{u} = 0, \\ \partial_t \rho + \boldsymbol{u}\cdot\boldsymbol{\nabla}\rho = (PrRe_L)^{-1}\nabla^2\rho, \end{cases} \tag{3.1}$$

with stress-free boundary conditions and no density flux at $z = 0, H$, and periodic boundary conditions in the horizontal. The forcing term $A\boldsymbol{u}$ is tuned at each time-step so as to have a bulk kinetic energy $\mathcal{E}_k = 1$ at all times, following the method of Yi & Koseff (2022). In particular, A satisfies

$$A = \frac{\epsilon + \mathcal{B} - \tau^{-1}(\mathcal{E}_k - 1)}{2\mathcal{E}_k}, \tag{3.2}$$

with $\mathcal{B}$ the buoyancy flux. This ensures that a constant kinetic energy $\mathcal{E}_k = 1$ is reached within τ units of time. Here, $\tau = 0.1$ (in dimensionless, simulation time units).

We initialise the simulation with a linearly stratified density field with dimensional buoyancy frequency $\hat{N}$ and a divergent-free, isotropic velocity with dimensional energy spectrum (Passot & Pouquet 1987)

$$\hat{E}(\hat{k}) = \frac{32\hat{E}_0}{\hat{k}_0}\sqrt{\frac{2}{\pi}}\left(\frac{\hat{k}}{\hat{k}_0}\right)^4 \exp\left[-2\left(\frac{\hat{k}}{\hat{k}_0}\right)^2\right]. \tag{3.3}$$

The initial kinetic energy $\hat{E}_0$ and wavenumber with maximum energy $\hat{k}_0$ determine a velocity and length scale $\hat{U} := \sqrt{\hat{E}_0}$ and $\hat{L} := 1/\hat{k}_0$, respectively. We use the dynamic pressure $\hat{\rho}_0\hat{U}^2$ to scale pressure and $\hat{L}\hat{\rho}_0\hat{N}^2/\hat{g}$ to scale density (with $\hat{\rho}_0$ a reference density). Time is scaled advectively so that $Re_L := \hat{U}\hat{L}/\hat{\nu}$ and $Ri_L := \hat{N}^2\hat{L}^2/\hat{U}^2$. The dimensionless horizontal box size is 10×10. Simulations are run until the density field is homogeneous. We fix $Re_L = 100$, and choose the value of Ri_L and $H := \hat{H}/\hat{L}$ to target specific values of $Ri = \hat{N}^2\hat{H}^2/\hat{\mathcal{E}}_k = Ri_L H^2/\mathcal{E}_k$ (with $\hat{\mathcal{E}}_k$ scaled by $\hat{U}^2$). The value of $\hat{\epsilon}$ and, hence, $t_M/t_R \sim \hat{\nu}\hat{N}^2\hat{\epsilon}^{-1}\mathcal{F}(Pr) = Ri_L\langle\partial_i u_j \partial_i u_j\rangle_V^{-1}\mathcal{F}(Pr)$ (with $\langle\cdot\rangle_V$ being a volume average; here, $\hat{\epsilon}$ is scaled by $\hat{\nu}\hat{U}^2/\hat{L}^2$) arises as a property of the flow, and is in fact time dependent (see figures 2*b* and 2*c*) since no statistical equilibrium can be reached as no background stratification is imposed. More precisely, as velocity gradients develop in the flow, ϵ grows. When the density field is fully mixed and $\mathcal{B} = 0$, the value of ϵ is essentially fixed by the forcing parameter A (see (3.2)). The forcing chooses to destroy small-scale velocity fluctuations and drives the water column at just the right speed to have $\mathcal{E}_k = 1$. As a result, ϵ decreases (see figure 2*b*). We will assume, admittedly somewhat cavalierly, that the parameter t_M/t_R is well characterised by its time average. We fix $Pr = 1$ so that $\mathcal{F}(Pr) \simeq 1$. The simulation parameters are summarised in figure 2.

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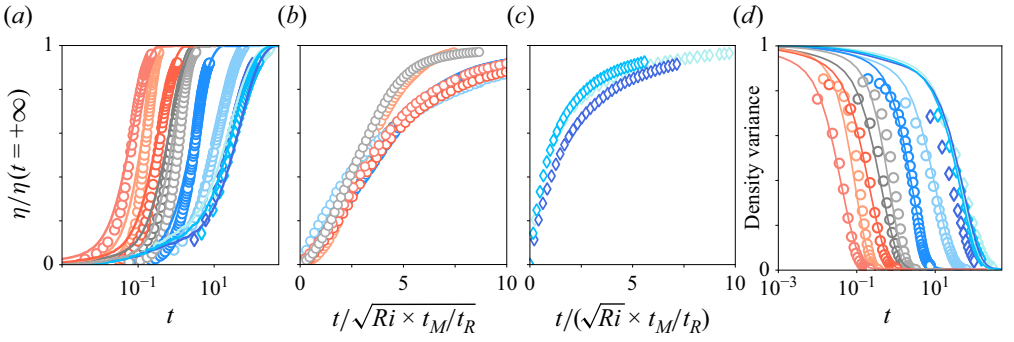


Figure 3. (*a*) Time evolution of the cumulative mixing efficiency η (symbols) with prediction from model (2.6) (solid lines). In panels (*b*) and (*c*), time has been rescaled using predictions of § 2 in the 'fast' mixing (O) and 'slow mixing' (◇) regimes. (*d*) Same as panel (*a*) for the time-evolution of the density variance, scaled by its initial value.

The time evolutions of η are presented in figure 3 with the corresponding prediction from the model (2.6) (numerically solved using 20 depth levels and 100 density levels). Note that the simulation time is multiplied by $Ri_L\langle\partial_i u_j\partial_i u_j\rangle_V^{-1/2}$ to be consistent with the dimensionless time used in (2.6). We found that the best fits occur for $t_M/t_R \simeq 5.5\hat{\nu}\hat{N}^2\hat{\epsilon}^{-1} = 5.5/Re_b$. When time is rescaled by the algebraic combination of Ri and t_M/t_R suggested in § 2, the data collapse on a single curve, suggesting that these times indeed control the evolution of the density field from its initial to a fully mixed state.

Another important measure of mixing is the time evolution of the density histogram variance, characterising the diversity of density levels still present in the flow as it mixes. In figure 3(*d*), we present the time evolution of the density variance as it decreases from its initial value (characterising the spread of density levels present in the initial configuration of the density field) to 0 (i.e. fully homogeneous density field, no variance). Again, the model (2.6) appears to capture the simulation trends, suggesting that the modelling assumptions used in this work are relevant to describe mixing.

4. Discussion

Using heuristic arguments (and sometimes brusque approximations and assumptions), we proposed different time scales that govern the evolution of a density field as it is stirred and mixed towards homogeneity. The time scales have been shown to depend on the amount of energy that stirs the field, its dissipation rate, as well as a measure of the stratification against which turbulence has to work and the molecular properties of the scalar being mixed. These time scales can potentially be used in coarse-resolution models to dynamically constrain the relaxation time of the density field towards its mean value (provided accurate sub-grid estimates of ϵ and $\mathcal{E}_k$ are accessible). More generally, we have shown that the mixing history and routes towards homogeneity in stratified turbulent flows is controlled by the competing effects of stirring, stretching-enhanced molecular diffusion and restratification. Whereas kinematic stretching of fluid parcels enhances molecular mixing, buoyancy forces hampers mixing by restratifying the density field and limiting the density levels that fluid parcels encounter along their stochastic paths and hence mix into. Modelling mixing as a convolution operation acting on the mixing time scale t_M, we developed a simple model for the evolution of the probability of finding a density level at a given depth as the water column is stirred and irreversibly mixed.

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It is important to note that the scalings for L_S/H, t_M/t_R and, hence, of the model parameters depend on the initial density field configuration. This raises the following question: as a strongly linearly stratified flow evolves into a layered state, as experimentally observed by Park, Whitehead & Gnanadeskian (1994) for instance, which scaling should be used? This raises the question of whether the parameters of the model are functions of bulk or local quantities. For instance, we assumed that D was a function of the initial bulk density stratification even after the initial density field was mixed. It seems perhaps more realistic to consider D as a function of the instanteneous mean density gradient $\partial_z\overline{\rho}$ and consider a model of the form

$$\partial_t p = \partial_z \left[D(\partial_z\overline{\rho})\partial_z p + \frac{D(\partial_z\overline{\rho})g}{\beta\rho_0}(\rho - \overline{\rho})p \right] + \text{mixing terms}. \tag{4.1}$$

Multiplying by ρ and integrating, we obtain an equation for the mean density profile $\overline{\rho}$ of the form

$$\partial_t \overline{\rho} = \partial_z \left[D(\partial_z\overline{\rho})\partial_z\overline{\rho} + \frac{D(\partial_z\overline{\rho})g}{\beta\rho_0}\left(\overline{\rho^2} - \overline{\rho}^2\right) \right] + \text{mixing terms}. \tag{4.2}$$

This equation, which accounts for the stochastic reorganisation of the density field, can be prone to an anti-diffusive instability provided the flux $D(\partial_z\overline{\rho})\partial_z\overline{\rho}$ is a decreasing function of $\partial_z\overline{\rho}$. This is reminiscent of the mechanism hypothesised by Phillips (1972) to explain layer formation in stratified turbulent flows. Is it the case? Is layering encoded in model (2.5), provided D is allowed to vary with the local density gradient $\partial_z\overline{\rho}$ and hence the density level histogram p? This question, outside of the scope of the present work, is left for future investigation. This is however, a crucial one; layered states are key features of the strongly stratified or layered anisotropic stratified regimes (Riley & Lindborg 2008; Falder, White & Caulfield 2016) relevant to geophysical flows. How should the model presented here be adapted to take into account the strong intermittencies and anisotropies characteristic of these regimes?

Finally, is the $Ri < 1, t_M/t_R > Ri$ regime accessible? Indeed, it corresponds to a weakly stratified regime in terms of energetics, but a strongly stratified one in terms of mixing (vertically disturbed fluid parcels return to their equilibrium position before having time to mix). In fact, it seems (see figure 2, for instance) that there exists a (loose) relationship between Ri and t_M/t_R (at least for flows considered here), a result that would enable the description of mixing in stratified turbulent flows using a single parameter.

Acknowledgements. Discussions with Jeff Koseff are gratefully acknowledged.

Funding. The support of ONR to CTR under grant N000142312833 is gratefully acknowledged.

Declaration of interests. The author reports no conflict of interest.

Appendix A. Conservation properties and steady-state solution of the model

To be physically relevant, the model (2.6) with its boundary condition should have the following conservation properties. We set $s = 1$ here.

(i) Normalisation. We want that the probability of finding a fluid parcel at each depth is 1, i.e.

$$\forall z \in [0, 1], \quad F[z] := \int p \, \mathrm{d}\rho = 1. \tag{A1}$$

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Intergrating (2.6) with respect to density levels ρ gives, for all z (using Fubini theorem and changes of variables to simplify the convolution integral),

$$\partial_t F = Ri^{-1}\partial_z^2 F + Ri^{-1/2}\partial_z[\overline{\rho}(1-F)] - \frac{t_R}{t_M}\big(-F+F^2\big). \tag{A2}$$

Then, if the normalisation constraint is satisfied initially, it will be at all subsequent times.

(ii) Mass conservation. We want the overall mass of the system to be conserved, i.e.

$$\langle\rho\rangle := \int \rho \int p \,\mathrm{d}z\mathrm{d}\rho \tag{A3}$$

independent of time. Multiplying (2.6) by ρ and integrating with respect to the density levels ρ gives the following equation for the mean density at each depth $\overline{\rho}$:

$$\begin{aligned}\partial_t\overline{\rho} = {} & Ri^{-1}\partial_z^2\overline{\rho} + Ri^{-1/2}\partial_z\big[\overline{\rho^2}-\overline{\rho}^2\big] \\ & + \frac{t_R}{t_M}\left[\overline{\rho} - \frac{1}{\alpha(1-\alpha)}\int\rho\int p\left(\frac{s}{\alpha}\right)p\left(\frac{\rho-s}{1-\alpha}\right)\mathrm{d}s\mathrm{d}\rho\right].\end{aligned} \tag{A4}$$

Here, $\overline{\rho^2} = \int \rho^2 p\,\mathrm{d}\rho$. Using the fact that $\rho = s + (\rho - s)$ and changes of variables, the last term becomes $\alpha\overline{\rho} + (1-\alpha)\overline{\rho} = \overline{\rho}$. Summarising,

$$\partial_t\overline{\rho} = Ri^{-1}\partial_z^2\overline{\rho} + Ri^{-1/2}\partial_z\big[\overline{\rho^2}-\overline{\rho}^2\big]. \tag{A5}$$

This equation shows that the convolution operation does not alter the local mean density profile; mass is reorganised by stirring. Integrating with respect to depth z, we get

$$\partial_t\langle\rho\rangle = \big[Ri^{-1}\partial_z\overline{\rho} - Ri^{-1/2}\big(\overline{\rho^2}-\overline{\rho}^2\big)\big]_{z=0}^{z=1}. \tag{A6}$$

The left-hand side vanishes because of the boundary conditions and, hence, mass is conserved.

(iii) Steady-state solution. The model should have the Dirac distribution $\delta(\rho - \langle\rho\rangle)$ (at each depth z) as the steady-state solution. This solution corresponds to the fully mixed state. Since this distribution does not depend on z, it suffices to show that it satisfies $-p + p_\alpha * p_{1-\alpha} = 0$. In fact, Dirac distributions are the only solutions of that equation. Indeed, denoting $\hat{p}(s)$ the Laplace transform of $p(\rho)$, the above is equivalent to $\hat{p}(s) = \hat{p}(\alpha s)\hat{p}((1-\alpha)s)$ or, with $f = \log(\hat{p}) = \sum f_n s^n$,

$$\forall n, \quad f_n = f_n\alpha^n + f_n(1-\alpha)^n. \tag{A7}$$

This can be satisfied for $n = 1$ only. Hence, $f(s) = f_1 s$. Therefore, $\hat{p}(s) = \mathrm{e}^{f_1 s}$ so that $p(\rho) = \delta(\rho + f_1)$. By conservation of the mass, f_1 as to be $\langle\rho\rangle$ and, hence, we have the desired result.

References

Batchelor, G.K. 1959 Small-scale variation of convected quantities like temperature in turbulent fluid Part 1. General discussion and the case of small conductivity. *J. Fluid Mech.* **5** (1), 113–133.

Burns, K.J., Vasil, G.M., Oishi, J.S., Lecoanet, D. & Brown, B.P. 2020 Dedalus: a flexible framework for numerical simulations with spectral methods. *Phys. Rev. Res.* **2** (2), 023068.

Caulfield, C.P. 2021 Layering, instabilities, and mixing in turbulent stratified flows. *Annu. Rev. Fluid Mech.* **53** (1), 113–145.

Cimoli, L., *et al.* 2023 Significance of diapycnal mixing within the Atlantic meridional overturning circulation. *AGU Adv.* **4** (2), e2022AV000800.

https://doi.org/10.1017/jfm.2026.11411 Published online by Cambridge University Press

DUPLAT, J., INNOCENTI, C. & VILLERMAUX, E. 2010 A nonsequential turbulent mixing process. *Phys. Fluids* **22** (3), 035104.

FALDER, M., WHITE, N.J. & CAULFIELD, C.P. 2016 Seismic imaging of rapid onset of stratified turbulence in the South Atlantic Ocean. *J. Phys. Oceanogr.* **46**, 1023–1044.

KIMURA, Y. & HERRING, J.R. 1996 Diffusion in stably stratified turbulence. *J. Fluid Mech.* **328**, 253–269.

MARRA, J., BIDIGARE, R.R. & DICKEY, T.D. 1990 Nutrients and mixing, chlorophyll and phytoplankton growth. *Deep Sea Res. Part A. Oceanogr. Res. Pap.* **37** (1), 127–143.

PARK, Y., WHITEHEAD, J.A. & GNANADESKIAN, A. 1994 Turbulent mixing in stratified fluids: layer formation and energetics. *J. Fluid Mech.* **279**, 279–311.

PASSOT, T. & POUQUET, A. 1987 Numerical simulation of compressible homogeneous flows in the turbulent regime. *J. Fluid Mech.* **181**, 441–466.

PETROPOULOS, N., DE BRUYN KOPS, S.M. & CAULFIELD, C.P. 2025 Modelling dispersion in stratified turbulent flows as a resetting process. *J. Fluid Mech.* **1002**, A48.

PETROPOULOS, N., CAULFIELD, C.P., MEUNIER, P. & VILLERMAUX, E. 2023 Settling versus mixing in stratified shear flows. *J. Fluid Mech.* **974**, R1.

PHILLIPS, O.M. 1972 Turbulence in a strongly stratified fluid—Is it unstable? *Deep Sea Res. Oceanogr. Abstr.* **19**, 79–81.

RILEY, J.J. & LINDBORG, E. 2008 Stratified turbulence: a possible interpretation of some geophysical turbulence measurements. *J. Atmos. Sci.* **65** (7), 2416–2424.

TSENG, Y. & FERZIGER, J.H. 2001 Mixing and available potential energy in stratified flows. *Phys. Fluids* **13** (5), 1281–1293.

VENAILLE, A., GOSTIAUX, L. & SOMMERIA, J. 2017 A statistical mechanics approach to mixing in stratified fluids. *J. Fluid Mech.* **810**, 554–583.

VILLERMAUX, E. 2019 Mixing versus stirring. *Annu. Rev. Fluid Mech.* **51**, 245–273.

WUNSCH, C. & FERRARI, R. 2004 Vertical mixing, energy, and the general circulation of the oceans. *Annu. Rev. Fluid Mech.* **36** (1), 281–314.

YI, Y.R. & KOSEFF, J.R. 2022 Dynamics and energetics underlying mixing efficiency in homogeneous stably stratified turbulence. *Phys. Rev. Fluids* **7** (8), 084801.

https://doi.org/10.1017/jfm.2026.11411 Published online by Cambridge University Press